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An evaluation and risk assessment of children's exposures to water-soluble per- and polyfluoroalkyl substances through winter gloves

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ABSTRACT

Winter gloves are often treated with fluoroacrylic surface coatings containing per- and polyfluoroalkyl substances (PFAS) to enhance water resistance. Concern over exposure to water-soluble PFAS, those with the greatest toxicological and regulatory relevance, has grown, particularly for children, who may experience higher relative body burdens due to lower body weight and frequent hand-to-mouth behaviors. In this study, we characterized the PFAS content and migration potential of winter gloves and conducted a screening-level risk assessment of perfluorooctanoic acid (PFOA) and perfluorohexanoic acid (PFHxA) glove exposures in children aged 2–6, considering both hand-to-mouth transfer and dermal absorption. Experimental data were generated through total fluorine analysis, liquid chromatography–tandem mass spectrometry (LC-MS/MS), gas chromatography–tandem mass spectrometry (GC-MS/MS), and leachate testing of individual glove components across eight glove brands. Three fluorotelomer compounds were detected above the limits of detection in 6 out of 173 components, spanning 3 of the 8 brands. No PFAS were detected in leachate samples, suggesting negligible migration under simulated use conditions. The calculated oral, dermal, and cumulative hazard indices, based on conservative, low-exposure scenarios using estimated concentrations below the detection limit, were all well below 0.001. This study contributes to the evidence base for PFAS risk assessment in consumer products and demonstrates that detectable PFAS residues do not necessarily translate to meaningful exposure or health risk. These findings support the use of risk-based regulatory approaches that incorporate realistic exposure scenarios in the evaluation of PFAS in treated textiles.

1. Introduction

Per- and polyfluoroalkyl substances (PFAS) are widely used in waterproof textiles due to their hydrophobic and stain-resistant properties [1]. PFAS are broadly defined as a class of synthetic fluorinated compounds characterized by strong carbon-fluorine bonds. Amidst the wide range of PFAS, the perfluoroalkyl acids (PFAAs) are of the greatest regulatory interest owing to their purported water solubility, bioaccumulation potential, environmental persistence, and toxicity [2,3]. These terminal PFAS are characterized by fluorinated alkyl chains with either carboxylate or sulfonate functional groups [3].

Concerns regarding PFAA exposure from clothing have increased as research continues to highlight their historical and current use as water-resistant coatings for fabrics. Fluorotelomer alcohols are starting chemicals and intermediate degradation by-products in the production of the majority of commercial PFAS, including fluorotelomer-based polymers [4]. Their presence is an indication of treatment with PFAS even if the identity of the PFAS is not known [4].

Fluorotelomers are often polymerized onto an acrylic backbone as part of the surface coatings used in water-resistant textiles [5]. They are typically incorporated via relatively weak ester moieties, which may result in their degradation and potential migration from the polymer wherein they may further degrade into terminal PFAS [5]. Additionally, small amounts of unreacted PFAS salts can remain on the textile surface as residuals from the manufacturing process [6]. Two of the most common fluorotelomers, 8:2 FTOH and 6:2 FTOH, have been reported to degrade into perfluorooctanoic acid (PFOA) and perfluorohexanoic acid (PFHxA) both by metabolism and in the environment [7–9].

The regulatory landscape for PFAS in consumer products is evolving rapidly, particularly in North America. Several U.S. states have proposed eliminating the use of PFAS in textiles intended for extreme weather conditions and high-performance applications in occupational settings

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[10,11]. In some cases the essential-use concept has been introduced as a framework to guide the phase-out of PFAS by assessing whether a given use is truly essential, based on three criteria: the function provided, its necessity for health, safety, or critical societal functions, and the availability of technically and economically feasible alternatives [12].

Given the increasing concern surrounding PFAS presence in clothing and other textiles, it is useful to thoroughly characterize potential human exposure from treated fabrics. Even if human exposure to textiles via clothing is deemed safe, because the PFAS will normally enter the water via washing of the clothing, bans due to their environmental persistence seem possible.

This study quantifies PFAS in water-resistant gloves, evaluates their potential for detachment and migration from polymerized surface coatings under typical use conditions, and presents a risk assessment for identifying risk to human health.

2. Methods

2.1. Sampling and analytical techniques

A total of 13–36 distinct glove components per glove were dismantled and assayed for PFAS across eight brands of American and Canadian-made gloves. These components included brand tags, inner linings, exterior palm and back fabrics, clasps, straps, cuffs, reflective strips, and other structural elements. Sampling excluded impermanent tags such as pricing or sizing stickers. Table 1 summarizes the glove sampling design.

Each glove was mechanically dismantled into its individual components following a modified version of IEC 62321-2 (Images 1 and 2). IEC 62321-2 is recognized as a comprehensive disassembly, disjointment, and mechanical sample preparation standard for dismantling complex articles into homogeneous material fractions. For the purposes of this assessment, the internal protocol was modified to accommodate coated textile products by treating each side of a coated fabric material as a discrete analytical layer.

The dismantling process was further aligned with the analytical guidelines of IEC 62321-3-1, which outlines standardized procedures for X-ray fluorescence (XRF) analysis of hazardous substances in non-homogenous materials. Following this protocol, total fluorine analyses were conducted using a Bruker® S8 TIGER Series 2 spectrometer, a high-powered wavelength-dispersive XRF (WD-XRF) instrument designed for elemental analysis. Fluorine was quantified using its Kα1 emission line at 0.677 keV, with corrections applied for potential spectral interference from the Lα1 emission line of iron (0.705 keV) and the Mα1 emission line of cerium (883 keV). The instrument's vacuum chamber enables accurate measurement of low atomic number elements such as fluorine by minimizing atmospheric attenuation of low-energy X-ray emissions. The method detection limit for elemental fluorine within fluoropolymers on the sample surface was 100 ppm.

Glove components exhibiting detectable fluorine concentrations via WD-XRF were subjected to targeted PFAS analysis in accordance with EN 17681-1:2022 and EN 17681-2:2022. These standards specify liquid chromatography–tandem mass spectrometry (LC-MS/MS) and gas chromatography–tandem mass spectrometry (GC-MS/MS) methods, respectively, for the identification and quantification of extractable perfluorinated and polyfluorinated compounds in textile materials and coated fabrics. Selection of analytes for either LC-MS/MS or GC-MS/MS analysis was guided by chemical properties such as volatility, polarity, and molecular weight, which influence method suitability.

Table 1
Number of distinct glove components analyzed across each of eight winter glove brands.

Glove	Number of distinct glove components reported
A	17
B	15
C	36
D	24
E	22
F	15
G	31
H	13



Image 1. Sample nomenclature for distinct glove components (glove G).

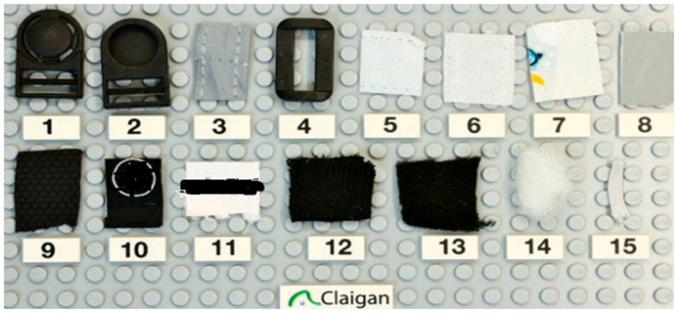


Image 2. Representative image of glove components extracted from one brand of gloves selected for the study (glove B).

A total of 38 analytes were evaluated: 25 perfluoroalkyl carboxylates (PFCAs) and 13 perfluoroalkyl sulfonates (PFSAs). Extractions were performed using 14 mL of LC-MS-grade methanol and 1.4 mL of a 40 % aqueous sodium hydroxide (NaOH) solution. The samples underwent ultrasonic-assisted extraction at 60 ± 5 °C for 60 min. Post-extraction, the pH of each solution was adjusted to 6.0–7.0 using glacial acetic acid prior to instrumental analysis.

Migration testing of glove components with detectable targeted PFAS concentrations followed a modified EN 1811 protocol, a method traditionally used to assess dermal transfer of metals following prolonged skin contact. This approach was adapted to estimate PFAS migration, providing a relevant measure of potential human exposure. Samples were immersed in 6 % artificial sebum for 24 h to mimic PFAS solubilization into human skin lipids on the skin surface, and the resulting fluid was analyzed by LC-MS/MS. Leachate testing was performed with artificial sebum rather than artificial saliva because human sebum is thought to be a more aggressive leachate due to its chemical composition and acidity. Artificial sebum is widely used in leaching tests for evaluating consumer product safety.

3. Sampling results

Total fluorine analysis detected measurable fluorine concentrations in seven of the eight gloves tested, with values ranging from 72 ± 10 ppm to 10,583 ± 72 ppm (Table 2).

PFAS-specific analysis using LC-MS/MS identified detectable levels of PFAS in three of the eight gloves (Table 3). Among the 173 individual glove components analyzed, only six components tested positive for PFAS across the three gloves. These detections were either fluorotelomer alcohols or fluorotelomer acrylates and no terminal PFAS were detected.

On average, solvent-extracted PFAS concentrations accounted for approximately 0.23 % of the total fluorine content in gloves with detections with maximum concentrations of 0.013 ppm for 8:2 fluorotelomer alcohol (8:2 FTOH), 0.014 ppm for 6:2 fluorotelomer acrylate (6:2 FTA), and 0.29 ppm for 6:2 fluorotelomer alcohol (6:2 FTOH) (Table 3). The most frequently detected PFAS was 6:2 FTOH, which was identified in five glove components across the three different brands.

These results differ substantially from the glove testing data generated by Claigan during the winter of 2024, which involved the same dismantling and analytical procedures (albeit not necessarily the same glove types). Eight out of eleven gloves tested contained some form of PFAS, many of which were PFAAs (Table 4).

Notably, migration testing in the current study did not detect any PFAS in glove components that initially tested positive for measurable PFAS concentrations.

4. Risk assessment

A human health risk assessment was conducted to evaluate non-cancer adverse effects associated with oral and dermal exposure to PFOA and PFHxA in children aged 2 to <3 years and 3 to <6 years resulting from the use of winter gloves.

The two compounds of interest were evaluated as potential degradation products of 8:2 FTOH, 6:2 FTOH, and 6:2 FTA, each of which was detected in at least one solvent extraction analysis. These degradation products were included in the risk assessment calculations in lieu of their precursors due to both regulatory concern and the availability of toxicological data to simplify hazard quotient calculations.

The oral and dermal routes were considered the most relevant pathways due to the nature of glove use, which involves prolonged skin contact and the potential for hand-to-mouth activity in young children. The selected age cohorts reflect the populations most likely to wear gloves for extended durations and engage in frequent mouthing behaviors, increasing their susceptibility to exposure. Furthermore, children may experience greater relative internal doses compared to adults, supporting a health-protective assumption that evaluating risks in children provides a conservative estimate for broader population protection.

For dermal exposure assessment, only glove components that come into direct contact with the skin were considered. For oral exposure assessment, only external glove materials with reasonable potential for direct contact with the mouth were included.

Any chemical concentrations present in gloves or other wearable textiles represent a finite and depletable mass. Accordingly, exponential decay models were applied to simulate declining chemical concentrations over time, under the assumption of steady absorption. These models assume that a consistent percentage of the remaining mass is

Table 2
Fluorine analysis results for all winter gloves tested.

Glove	Number of glove components with fluorine detects	Glove components with fluorine detects	Concentration (ppm)
A	1	Matte Exterior Dorsal Fabric	190 ± 11
B	2	Exterior Dorsal Fabric	190 ± 15
		Interior Tag	344 ± 12
C	8	Exterior Palmar Fabric A (Smooth)	90 ± 9
		Exterior Palmar Fabric B (Textured)	10,583 ± 72
		Exterior Dorsal Thumb Fabric	94 ± 10
		Exterior Palmar Thumb Fabric	1046 ± 17
		White Fingertip Insert	167 ± 22
		Black Fingertip Insert	1116 ± 26
		Thumb Lining Side A	605 ± 14
		Thumb Lining Side B	307 ± 12
D	1	Black Interior Band	463 ± 16
E	No Detect (ND)	ND	ND
F	2	Exterior Dorsal Fabric	202 ± 13
		Exterior Palmar Fabric	612 ± 16
G	4	Interior Cuff	184 ± 11
		Black Thread	68 ± 10
		Exterior Palmar Leather	72 ± 10
		Exterior Dorsal Glove Surface (Smooth)	258 ± 12
H	1	Exterior Palmar Fabric (Textured)	444 ± 13

Table 3
LC-MS/MS analysis of fluorine-positive components of all winter gloves tested.

Glove	Glove component tested	PFAS compound	Concentration (ppm)
A	Matte Exterior Dorsal Fabric	No Detect (ND)	ND
B	Exterior Dorsal Fabric	ND	ND
	Interior Tag	ND	ND
C	Exterior Palmar Fabric A (Smooth)	ND	ND
	Exterior Palmar Fabric B (Textured)	6:2 FTOH	0.29
	Exterior Dorsal Thumb Fabric	ND	ND
	Exterior Palmar Thumb Fabric	ND	ND
	White Fingertip Insert	ND	ND
	Black Fingertip Insert	6:2 FTOH	0.22
	Thumb Lining Side A	ND	ND
	Thumb Lining Side B	ND	ND
D	Black Interior Band	ND	ND
E	Not Applicable (N/A)	N/A	N/A
F	Exterior Dorsal Fabric	ND	ND
	Exterior Palmar Fabric	ND	ND
G	Interior Cuff	8:2 FTOH	0.013
	Black Thread	6:2 FTOH	0.038
	Exterior Palmar Leather	ND	ND
	Exterior Dorsal Glove Surface	6:2 FTA	0.014
	(Smooth)	6:2 FTOH	0.18
H	Exterior Palmar Fabric (Textured)	6:2 FTOH	0.038

Table 4
LC-MS/MS detections of water-soluble PFAS in winter gloves tested in 2024.

Glove	Glove component tested	PFAS compound	Concentration (ppm)
1: The North Face	Exterior Thumb Fabric	PFBA	0.01
		PFPeA	0.05
		PFHxA	0.03
		PFHpa	0.01
		4HPFUnA	0.03
2: Kombi	Fabric Loop Attachment	PFPeA	0.03
3: Lolmot	Textured Palm Fabric	PFOA	0.320
4: Kombi	Exterior Fabric	8:2 FTOH	0.058
5: Hot Paws	Yellow Brand Label	HPFHpA	0.02
6: Igloos	Green Fabric	8:2 FTOH	0.91
		10:2 FTOH	0.22
7: Columbia	Black Velvet Fabric	PFPeA	0.05
		PFHxA	0.03
		6:2 FTOH	3.54
8: Flapjack Kids	Outer Patterned Fabric	PFOA	0.02
		PFPeA	0.03
		PFHxA	0.02
		8:2 FTOH	0.10

absorbed during each exposure event, a practical approach for estimating chronic exposure in the absence of compound-specific depletion or release data.

Chronic oral and dermal intake values of each of the two PFAS compounds of interest, for both age groups, were calculated using the equations and pathway-specific assumptions described below.

4.1. Oral exposure

Equation 1. *PFAS remaining following oral ingestion*

$$C_{n-oral} = C_0 \quad * (1 - r)^n$$

Where:

C_{n-oral} = Concentration remaining (mg)
C_0 = Initial concentration (mg)
r = Depletion rate (fraction/contact)
n = Contact frequency (contacts)

4.1.1. Initial concentration

Concentrations from migration testing were used for exposure assessment, as they more accurately represent the fraction of PFAS that transfers to the skin surface or comes into contact with the mouth in a worst-case scenario. However, since no fluorotelomer or fluorotelomer acrylate concentrations were detected above the limit of detection (LOD), a theoretical approach was applied by assigning a value equal to one-half (½) the LOD of 10 ppb (converted to mass in mg) for each target analyte to account for potential low-level presence in exposure calculations [13,14].

In addition, because PFOA and PFHxA were assessed as breakdown products of detected fluorotelomer alcohols, concentrations equivalent to ½ the LOD were further fractionated in accordance with biological breakdown percentages of these alcohols into terminal PFAS in the human body.

8:2 FTOH undergoes biotransformation into PFOA and perfluorononanoic acid (PFNA) through multiple oxidative steps, enzymatic interactions, and potential β-oxidation-like processes [9]. The relative formation of PFOA versus PFNA depends on species-specific metabolic pathways. In humans, 8:2 FTOH preferentially degrades into PFOA due to the limited efficiency of enzymatic elongation required for PFNA formation and the inherent tendency of β-oxidation to favor shorter-chain metabolites [9]. However, Nabb, Szostek [9] demonstrated that the overall metabolism of 8:2 FTOH in humans is slow and inefficient, with only 0.02 % of the parent compound converting to PFOA within two hours. They further estimated the half-life of 8:2 FTOH in human hepatocytes to be 35.9 min, indicating near-complete elimination at two hours. We assessed the metabolism of 8:2 FTOH into PFOA based on these findings in our risk assessment.

Similarly, 6:2 FTOH may be metabolized into PFHxA, perfluorobutanoic acid (PFBA), and perfluoroheptanoic acid (PFHpA) in biological systems along with 6:2 FTA, which is an intermediate degradation product [7]. Due to the limited availability of metabolic breakdown data for 6:2 FTA in humans, we conservatively applied the breakdown rates of 6:2 FTOH to both compounds as a surrogate. Among the possible PFCA metabolites, PFHxA was selected as the primary analyte for assessment due to stronger evidence of its formation in humans and its higher concentrations observed in the Russell, Himmelstein [7] study [8].

Data on the human toxicokinetics of 6:2 FTOH remain scarce. Studies have demonstrated that the majority of 6:2 FTOH undergoes conjugation reactions, forming glucuronide and glutathione derivatives rather than being directly converted to PFCAs [7,15]. Russell, Himmelstein [7] summarized findings from a previously conducted study by the same authors, reporting that the data from an experiment involving inhalation exposures of 1, 5, and 23 days as well as a 90-day oral exposure period in rats, revealed that a maximum of 1.9 % of administered 6:2 FTOH was metabolized into PFHxA. Given these findings, we conservatively assumed near-total elimination of the parent compound within 24 h. We adopted 2 % as an upper-bound estimate for our risk assessment. Because human metabolism of fluorotelomer alcohols is generally slower and less efficient at producing PFCAs compared to rodents, this likely represents an overestimation of PFHxA formation in humans.

Concentrations were adjusted according to these percentages. For the final concentrations applied in the risk calculations, the initial mass was calculated as one-half the LOD (5 ppb) corresponding to 2.0×10^{-5} mg in the 4 mL testing solution, and then multiplied by 0.0002 for PFOA and by 0.02 for PFHxA to account for estimated biotransformation yields.

4.1.2. Depletion rate

The U.S. EPA's *Child Specific Exposure Scenarios Examples* report identifies an average saliva transfer efficiency of 16 % for hand-to-mouth contact based upon thumb sucking and finger mouthing behaviors [16].

4.1.3. Contact frequency

Based on professional judgment and climatic patterns in temperate regions, it was assumed that winter conditions persist for an average duration of 12 weeks per year. During this period, it was estimated that children in the 2 to <3 years and 3 to <6 years age groups wear gloves during both school recess and recreational activities on weekends. Daily exposure duration was conservatively estimated to be 60 min, reflecting an upper-bound scenario wherein children spend up to one hour each day of the week wearing water-resistant winter gloves.

To estimate the frequency of hand-to-mouth behaviors involving gloved hands, we relied on age-stratified contact rates derived from Weibull distributions reported in the U.S. Environmental Protection Agency (EPA) [17] *Child-Specific Exposure Factors Handbook*. According to Xue, Zartarian [18], children aged 2 to <3 years exhibit an average of 5.3 hand-to-mouth contacts per hour in outdoor settings (rounded to 5 for this assessment), while children aged 3 to <6 years exhibit an average of 8.5 contacts per hour (rounded to 9).

The contact frequency represents the total number of age-specific hand-to-mouth contact events occurring over the 12 week annual exposure period.

Equation 2. *Total oral intake*

$$AB_{Oral} = C_0 - C_{n-oral}$$

Where:

AB_{Oral} = Absorbed dose (mg)
C_0 = Initial concentration (mg)
C_{n-oral} = Concentration remaining (mg)

Equation 3. *Oral Average Daily dose*

$$ADD_{Oral} = \frac{AB_{oral}}{BW \times AT}$$

Where:

AB_{oral} = Absorbed dose (mg)
BW = Body weight (kg)
AT = Averaging time (days)

4.1.4. Body weight

Mean body weights for both age groups were gathered from Chapter 8, Table 8–3 of the U.S. Environmental Protection Agency (EPA) [17] *Child-Specific Exposure Factors Handbook*.

4.1.5. Averaging time

A conservative estimate of one year (365 days) was used, based on the assumption that children in the evaluated age groups are likely to outgrow their winter gloves annually.

4.2. Dermal exposure

For parameters common to both the oral and dermal exposure equations, identical values were applied when applicable.

Equation 4. *PFAS remaining following dermal absorption*

$$C_{n-dermal} = C_0 \quad * (1 - r)^n$$

Where:

$C_{n-dermal}$ = Concentration remaining (mg)
C_0 = Initial concentration (mg)
r = Depletion rate (fraction/contact)
n = Contact frequency (contacts)

4.2.1. Depletion rate

Dermal absorption percentages for PFOA and PFHxA were derived from existing literature.

Abraham and Monien [19] conducted the only known *in vivo* dermal absorption study of PFOA in a human subject, applying PFOA-spiked sunscreen to the skin. The study used an applied dose of 0.0055 μg/cm^2 and monitored serum levels over a 115 day period, parameters that offer direct relevance to the current assessment. Based on these data, the authors estimated an average cumulative absorption of 1.6 %, which was adopted in this assessment as a conservative estimate of PFOA absorption per dermal glove contact.

In the absence of *in vivo* human data for PFHxA, a per-contact absorption rate of 5 % was selected based on findings from Hall, Baynes [20], who evaluated dermal uptake using artificial sweat on occluded porcine skin. The use of artificial sweat was considered relevant for this assessment, as was the application of occlusion. In addition, porcine skin is widely recognized as an appropriate surrogate for human skin in *in vitro* dermal absorption studies due to its structural and barrier similarities [21,22].

4.2.2. Contact frequency

Absorption was considered on a per-contact basis, and each glove use was assumed to represent one hour of continuous dermal contact over the 12 week annual exposure period.

Equation 5. *Total dermal absorption*

$$AB_{dermal} = C_0 - C_{n-dermal}$$

Where:

AB_{dermal} = Absorbed dose (mg)
C_0 = Initial concentration (mg)
$C_{n-dermal}$ = Concentration remaining (mg)

Equation 6. *Dermal Average Daily Dose*

$$ADD_{Dermal} = \frac{AB_{dermal}}{BW \times AT}$$

Where:

ADD_{Dermal} = Average daily dose (mg/kg-day)
AB_{dermal} = Absorbed dose (mg)
BW = Body weight (kg)
AT = Averaging time (days)

4.3. Hazard quotient (HQ) calculation

Final HQs were calculated using Equation 7 below.

Equation 7: *Individual and cumulative hazard quotients for oral and dermal exposures*

$$\text{CumulativeHQ} = \frac{\text{ADD}_{\text{Abs}}\text{ofPFAS}}{\text{RfDofPFAS}} + \frac{\text{ADD}_{\text{Oral}}\text{ofPFAS}}{\text{RfDofPFAS}}$$

Where:

$ADD_{Abs}of\ PFAS$ = Dermally absorbed average daily dose (mg/kg-day)
$ADD_{Oral}of\ PFAS$ = Oral average daily dose (mg/kg-day)
$RfD\ of\ PFAS$ = Chronic oral reference dose of PFAS (mg/kg-day)

Table 5
EPA chronic oral references doses (RfDs) for PFOA and PFHxA

Analyte	Reference Dose
PFOA	3.00×10^{-8}
PFHxA	5.00×10^{-4}

Table 6
HQs for children aged 2 to <3.

Analyte	PFOA	PFHxA	Cumulative HQ
Oral HQ	2.65×10^{-5}	1.59×10^{-7}	
Dermal HQ	1.96×10^{-5}	1.57×10^{-7}	4.64×10^{-5}
Total HQ	4.61×10^{-5}	3.16×10^{-7}	
Dermal/Oral HQ Ratios	1.35	1.01	

Table 7
HQs for children aged 3 to <6.

Analyte	PFOA	PFHxA	Cumulative HQ
Oral HQ	1.96×10^{-5}	1.18×10^{-7}	
Dermal HQ	1.46×10^{-5}	1.16×10^{-7}	3.44×10^{-5}
Total HQ	3.42×10^{-5}	2.34×10^{-7}	
Dermal/Oral HQ Ratios	1.35	1.01	

4.3.1. Reference dose

Chronic oral reference doses of PFOA and PFHxA published by the EPA for non-cancer endpoints were used to calculate hazard quotients (Table 5).

5. Risk assessment results

As shown in Tables 6 and 7 all individual and cumulative hazard quotients were well below 0.001; at least 100-fold below a dose of concern.

6. Discussion

Based on the available data, PFAS concentrations detected in winter gloves were minimal. When compared to the previous glove study, the current findings indicate a decline in both the frequency and diversity of PFAS detected in water-resistant textiles (Table 4). In the 2024 analysis, five distinct PFCAs were identified, whereas no PFCAs were detected in the present 2025 analysis. This apparent reduction is consistent with the broader trend of increasing regulatory restrictions and the industry's transition toward alternative, non-fluorinated water-resistant coatings.

The non-detect results from leachate testing suggest that the PFAS concentrations identified through solvent extraction do not readily migrate onto the skin under typical use conditions. This outcome highlights a key consideration: total fluorine and solvent-extracted PFAS concentrations alone are not reliable indicators of terminal, water-soluble PFAS available for dermal or oral uptake.

All calculated HQs for dermal and oral exposures were significantly below one under the exposure parameters assessed, indicating no significant non-cancer human health risk associated with PFAS exposures under the conditions assessed. Children in the age groups 2 to <3 and 3 to <6 are considered sensitive populations due to their higher relative chemical body burden, increased dermal absorption potential, longer durations of glove use, and higher frequencies of hand-to-mouth behaviors. The absence of risk in these populations indicates that PFAS exposure from winter glove use does not pose a health concern for the general population.

These results demonstrate that PFAS-containing water-resistant textiles can be worn without posing adverse health impacts under the exposure conditions assessed. Although a range of non-fluorinated durable water repellent (DWR) alternatives have been developed and marketed, they currently fall short of protection against low surface tension liquids which, to date, only short-chain fluorinated polymers have reliably provided [12]. This performance gap renders fluorinated textiles critical in certain occupational settings. For example, performance standards for surgical gowns, drapes, and clean air suits in the medical field require garments to withstand high-pressure liquid penetration by chemicals and bodily fluids while maintaining breathability during extended surgical procedures [12]. Similarly, firefighting gear must provide an equivalent combination of liquid repellency and breathability to protect emergency responders. Other industries, including oil and gas, as well as outdoor recreational sectors, also rely on high-performance textiles where such properties are needed [12].

While essential-use frameworks for PFAS appear practical in concept, their implementation remains uncertain due to legal, social, and economic challenges. The Toxic Substances Control Act (TSCA) polymer exemption has been criticized for its confusing eligibility criteria, vague definitions surrounding degradation and water absorption, strict structural exclusions, and burdensome recordkeeping requirements [23,24]. Additional factors, such as the potential for regrettable substitutions and the possibility of manufacturers withdrawing products from the market due to declining demand, further complicate regulatory implementation. These challenges could inadvertently restrict access to PFAS-based materials that are critical for essential uses [12,25].

6.1. Limitations and future research

It should be noted that PFAS-specific testing was limited to a targeted analyte list consisting of 25 PFCAs and 13 PFSAs (38 analytes total). Thus, the potential presence of other unmeasured PFAS compounds cannot be fully excluded.

This study did not assess the health hazards of volatile PFAS precursors that may be present in the atmosphere, particularly under storage conditions in poorly ventilated environments. Volatilization and subsequent exposure would likely depend on storage practices, proximity, and frequency of glove use by household members and is expected to be negligible in the present study. Nonetheless, future risk assessments should incorporate inhalation exposure models adapted for volatile PFAS species to better characterize this potential exposure route.

Given the low concentrations of PFAS detected, acute health effects are not expected. Furthermore, preliminary cancer risk estimates, calculated using the EPA's cancer slope factor of 0.07 mg/kg-day for PFOA, were de minimus [26].

7. Conclusions

- The findings of the current study indicate that the minimal concentrations of PFAS detected in winter gloves do not pose a risk to human health under the conditions assessed.
- Although the exposure assessment was theoretical in nature, it provides a conservative and scientifically defensible model for evaluating oral and dermal exposure risks associated with PFAS in textiles and similar consumer products.
- The methodological framework developed in this research could be applied to assessments of similarly manufactured clothing and protective gear.
- While public concern regarding PFAS remains substantial, the apparent lack of a human health risk in this study indicates that the performance benefits of PFAS-containing water-resistant textiles can be maintained without adverse health impacts.
- Regulatory decisions concerning PFAS in textiles should be informed by risk assessments that model real-world exposure scenarios using data on speciated terminal PFAS and their relevant precursors rather than by relying solely on measurements of total fluorine or solvent extraction content.

CRediT authorship contribution statement

Sarah Berlinski: Writing – review & editing, Writing – original draft, Visualization, Methodology, Investigation, Formal analysis, Conceptualization. **Bruce Calder:** Writing – review & editing, Resources, Methodology, Investigation. **Dennis Paustenbach:** Writing – review & editing, Visualization, Validation, Supervision.

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Declaration of Competing Interest

The authors declare the following financial interests/personal relationships which may be considered as potential competing interests: Two authors (Sarah Berlinski and Dennis Paustenbach) are employed by TRC, a consulting firm that provides scientific advice to the government, corporations, law firms, and various scientific/professional organizations. The firm has been engaged by manufacturers and distributors of PFAS and PFAS-containing compounds in various litigation matters and advisory roles. Dr. Dennis Paustenbach, who has been involved as an expert in many personal injury cases as an expert witness may be called upon in the future to serve as an expert in PFAS litigation. Bruce Calder declares that he has no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

Data will be made available on request.

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